

SafePay MENA

Idea Capture Template

MENA Ignite Open Gateway Hackathon - Phase 1 Submission

1. Submitter Details

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2. Idea Summary

2.1 Problem Statement

The MENA region's digital payment infrastructure is growing faster than the security systems protecting it.

The scale of the problem:

- Egypt's InstaPay processed **1.1 billion transactions** in H1 2025 (16M users). All transfers are instant and irreversible.
- UAE's Aani instant payment platform handles 25,000 P2P transfers daily across 12.5 million users.
- Saudi Arabia invested **\$469.9 million** in fraud detection in 2025 alone.
- The MENA digital payments market is projected to reach **\$275.47 billion by 2026**.

The Attack: SIM swap fraud is the primary vector. Fraudsters social-engineer telecom staff into transferring a victim's phone number to a new SIM, then intercept SMS OTP codes to authorize fraudulent bank transfers. Individual losses range from **\$270 to \$160,000+** per incident. A documented UAE case involved **\$1.5 million stolen** in a single attack. INTERPOL's Operation Ramz (2025-2026) arrested 200+ suspects across 13 MENA countries.

The root cause: Banks and telecom operators operate in completely separate data silos. When a SIM swap happens at Vodafone Egypt, no bank receives that signal. Telecoms have the fraud intelligence, but it has never been connected to payment systems in real-time. PwC's 2025 GCC report identified this as the critical 'fraud gap.'

The human cost: 64% of adults in the Arab region (~200 million people) remain unbanked. The #1 barrier is lack of trust. Every high-profile fraud case pushes millions further from formal financial systems.

2.2 Proposed Solution

SafePay MENA is an AI-powered fraud detection agent that acts as a real-time security layer between instant payment systems and the mobile network. Before any high-risk payment is authorized, the agent queries the telecom network using GSMA Open Gateway CAMARA APIs to verify the sender's identity and device integrity at the carrier level.

How it works:

1. User initiates a payment via InstaPay, Aani, or stc pay.
2. SafePay's AI agent receives transaction context (amount, sender phone, recipient, timestamp).
3. Agent autonomously decides which CAMARA APIs to query based on risk profile.
4. Agent queries SIM Swap status, Number Verification, Device Status, and Device Swap via Nokia NaC.
5. Signals are aggregated through a weighted risk scoring engine producing a score (0-100).
6. Agent decides: APPROVE (silent), STEP-UP authentication, or BLOCK with an alert.
7. Every decision includes a natural-language reasoning trace for regulatory audit compliance.

2.3 Expected Benefits

Stakeholder	Benefit
Consumers	Carrier-level security without behavior change. Eliminates interceptable SMS OTPs.
Banks & Fintechs	Access to telecom network fraud signals for first time. Reduced false positives. Audit trail.
Telecom Operators	New API monetization revenue stream. Demonstrated Open Gateway ROI.
MENA Region	Increased payment trust → accelerated financial inclusion for 200M unbanked Arabs.

3. Alignment

3.1 Challenge Theme: Theme 4: Secure Fintech, Payments & Anti-Fraud Innovation

3.2 GSMA Pillar: Industry Services and Solutions - Open Gateway

3.3 Regional Context: Egypt (InstaPay 1.1B tx), Saudi Arabia (SAMA \$470M fraud spend), UAE (Aani 12.5M users), Qatar (Ooredoo + Vodafone deployed CAMARA).

4. API Usage

SafePay integrates **4 CAMARA APIs** from the Nokia Network-as-Code platform:

#	API	Endpoint	Role	Weight
1	SIM Swap	POST /sim-swap/v0/check	Detects recent SIM changes	35%
2	Number Verification	POST /number-verification/v0/verify	Silent carrier auth (replaces OTP)	30%
3	Device Status	POST /device-status/v0/roaming	Detects roaming anomalies	15%
4	Device Swap	POST /device-swap/v0/check	Detects new handset change	20%

5. AI Agent Design

Built with **Google ADK + Gemini 2.0 Flash** following approved tooling guidelines. Agent dynamically plans tool calls, reasons across multiple signals, and outputs natural-language explanations.

6. Technical Architecture

Stack: FastAPI (Backend) + Google ADK (Agent) + Nokia NaC SDK (APIs) + Supabase (Database) + Next.js (Dashboard).

7. Business Model

- **Per-transaction check:** \$0.01 - \$0.05 per check (banks pay)
- **Monthly subscription:** \$500 - \$5,000/month for mid-size banks
- **Enterprise license:** \$50,000 - \$200,000/year for central banks
- **Addressable revenue:** Egypt InstaPay alone = \$44M/year at \$0.02/check.

8. Team Overview

Member	Role	Background
Karim Mohamed Abdelnabi	Solo Developer - Full Stack + AI	Full Stack & AI Developer with hands-on experience in Google ADK, Gemini API, Python, FastAPI, Supabase, and Next.js.

10. Criterion Alignment Summary

Criterion	SafePay Score Strategy
Relevance	Theme 4 alignment, Egypt/UAE/Saudi data, INTERPOL Ramz 200+ arrests across 13 countries.
Impact	\$275B market, 64% unbanked (200M people), 22 MENA markets, \$44M Egypt revenue potential.
Innovation	First multi-API AI agent orchestration layer. Replaces SMS OTP with carrier Number Verify.
Complexity	4 CAMARA APIs, Google ADK agent, weighted risk engine, FastAPI + Supabase + Next.js stack.